Pain

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Pain

Pain is defined as an unpleasant sensory associated with actual or potential tissue damage. It is the most common reason for seeking health care. Pain occurs as the result of many disorders, diagnostic tests, and treatments.

Importance of pain assessment and management

Pain management is considered such an important part of care that it is referred to as “the fifth vital sign” to emphasize its significance and to increase the awareness among health care professionals of the importance of effective pain management. Documentation of pain assessment is now as important as documentation of the “traditional” vital signs.

The pain care bill of rights

Pain Care Bill of Rights

As a person with pain, you have the right to:

- Have your report of pain taken seriously and be treated with dignity and respect by doctors, nurses, pharmacists, and other health care professionals.
- Have your pain thoroughly assessed and promptly treated.

- Be informed by your health care provider about what may be causing the pain, possible treatments, and the
- benefits, risks, and cost of each.
- Participate actively in decisions about how to manage your pain.
- Be referred to a pain specialist if your pain persists.

Components of pain

- A harmful stimulus which signals tissue damage.
- Personnel, private sensation of harm.

Types of pain according to origin

1. (superficial) pain usually involves the subcutaneous tissue. It produces sharp pain with a burning sensation.
2. Deep somatic pain: is diffused and originates in tendons, ligaments, bones, blood vessels, and nerves.

3. Visceral pain: is poorly localized and originates in body organs in the thorax, and abdomen.

4. Psychogenic pain: it is a pain where no physical pathology has been found or where the pain appears to have a greater psychological basis than a physiological one.

Classifications of pain by onset

1. ***Acute Pain***

It characterized by generally rapid in onset, varies in intensity from mild to severe, the cause is usually clear. It is usually easy to 'see' the pain, such as an injury or infection, it lasts few days or weeks until healing has occurred as acute pain from a broken leg. It may be associated with anxiety and fear. It consistently increases at night and during wound care, ambulation, coughing and deep breathing. If it is not effectively managed, it may progress to a chronic state.

2. Chronic Pain

It characterized by constant or intermittent and it interferes with normal functioning .It lasts for more than three months, or beyond normal healing time. It is a persistent pain that can disrupt sleep, mood and normal living. The cause is not always clear. Chronic pain may start with an injury or infection, or there may be an ongoing cause of pain (e.g. arthritis).It is associated with prolonged tissue pathology or it persists beyond the normal healing period for an acute injury or disease. It also increases at night.

3. Cancer-Related Pain

Pain associated with cancer may be acute or chronic. Pain in patients with cancer can be directly associated with the cancer (eg, nerve compression), a result of cancer treatment (eg, surgery or radiation).

Classification of pain by location

Pain can also be categorized according to location (eg, pelvic pain, chest pain). This type of categorization aids in communication about and treatment of the pain. For example, chest pain may suggest acute coronary syndrome (ACS).

Classification of pain by Etiology

Pain can also be categorized by etiology. As burn pain.

Manifestations of pain

1. **Physiologic responses:** observable physiological signs of acute pain include changes in blood pressure, heart rate, respiratory rate, increase blood pressure, diaphoresis, and pallor, muscle tension for body especially of face, nausea and vomiting if pain is severe.

2. Behavioral responses: -

- **Verbal responses:** they are the most dependable indicators of pain as crying.
- **Nonverbal responses:** they often give a clue about pain location. Common nonverbal responses include rubbing painful areas, restlessness, and clenched teeth.

Harmful Effects of Pain

- **Effects of Acute Pain**

(1)Unrelieved acute pain can affect the pulmonary, cardiovascular, gastrointestinal, endocrine, and immune systems.

(2)The widespread endocrine, immunologic, and inflammatory changes that occur with stress can have significant negative effects.

(3)Patients with severe pain and associated stress may be unable to take deep breaths and may experience increased fatigue and decreased mobility.

Effects of Chronic Pain

(1) Suppression of the immune function associated with chronic pain may promote tumor growth.

(2) Depression and disability.

(3) High dosages of opioid medications required to relieve chronic pain in some patients, can lead negative effects on other body systems as respiratory, renal and hepatic.

- **The neural mechanism by which pain is perceived involves a process that has four major steps:**
  - **Transduction**
  - **Transmission**
  - **Modulation**
  - **Perception**

1. **Transduction** is the conversion of a noxious stimulus (mechanical, chemical, or thermal) into electrical energy by a peripheral nociceptor (free afferent nerve ending). This is the first step in the pain process and can be inhibited by non- steroidal anti-inflammatory drugs (NSAID's), opioids and local anesthetics.

2.Transmission is part of a survival response to prevent tissue damage. A cycle of response to stimuli, followed by temporary hypersensitivity, then attenuation and resolution occurs.

3.Modulation Modulation of pain begins with an understanding of the various levels of pain modulation and extends to clinical interventions and protocols designed to reduce pain.

4.Perception is the process by which pain is recognized and interpreted by the brain.

Gate Control Theory

The classic gate control theory of pain, described by Melzack and Wall in 1965.

The gate control theory was the first theory to suggest that psychological factors play a role in the perception of pain. This theory helps explain how interventions such as distraction and music therapy relieve pain.

This theory proposed that stimulation of the skin evokes nervous impulses that are then transmitted by three systems located in the spinal cord. The substantia gelatinosa in the dorsal horn, the dorsal column fibers, and the central transmission cells.

Stimulation of the large-diameter fibers inhibits the transmission of pain, thus “closing the gate.” Conversely, when smaller fibers are stimulated, the gate is opened. This theory proposes a specialized system of large-diameter fibers.

Factors Influencing Pain Response

1. Past Experience

- It is tempting to expect that people who had multiple or prolonged experiences with pain will be less anxious and more tolerant of pain than those who have had little experience with pain.

2. Anxiety and Depression

- Although it is commonly believed that anxiety increases pain, this is not necessarily true. Anxiety that is related to the pain may increase the patient's perception of pain. Anxiety that is unrelated to the pain may distract the patient and may decrease the perception of pain.

3.Culture

Beliefs about pain and how to respond to it differ from one culture to the next. The nurse must react to the person's perception of pain.

4.Gender

The results of studies of gender in regard to pain levels and response to pain have been inconsistent. In some studies, women consistently reported higher pain intensity, frustration, and fear, compared to men.

The Nurse's Role in Assessment and Care of Patients with Pain

- **Assessment**

A broad definition of pain is, "whatever the person says it is, existing whenever the experiencing person says it does". This definition emphasizes the highly subjective nature of pain and pain management.

Although it is important to believe patients who report pain, it is equally important to be alert to patients who deny pain in situations where pain would be expected. Some people deny pain because they fear the treatment.

In assessing a patient with pain, the nurse reviews the patient's description of the pain and other factors that may influence pain, as well as the patient's response to pain relief strategies.

Characteristics of Pain

Pain assessment begins by careful patient observation. In addition, it is essential to ask the patient to describe, in his or her own words, the specifics of the pain.

Factors to be assessed in pain assessment

- **The factors to consider in a complete pain assessment are**: the intensity, timing, location, aggravating and alleviating factors.

1. Intensity

The intensity of pain ranges from none to mild discomfort to excruciating.

2. Timing

The nurse inquiries about the onset, duration, relationship between time and intensity (eg, at what time the pain is the worst). The nurse asks the patient if the pain began suddenly or increased gradually. Sudden pain that rapidly reaches maximum intensity is indicative of tissue injury, and immediate intervention is necessary.

3. Location

Having the patient point to the area of the body involved best determines the location of pain. Some general assessment forms include drawings of human figures, on which the patient is asked to shade in the area involved. This is especially helpful if the pain radiates (**referred pain**).

4. Quality

The nurse asks the patient to describe the pain in his or her own words. For example, the nurse asks the patient to describe what the pain feels like. The nurse must give the patient sufficient time to describe the pain, and the nurse must record all words in the answer. If the patient cannot describe the quality of the pain, the nurse can suggest words such as burning, aching, throbbing, or stabbing.

5. Personal Meaning

Pain means different things to different people; as a result, patients experience pain differently. It is important to ask how the pain affects the person's daily life. Some people with pain can continue to work or study, whereas others may be disabled by their pain, thus affecting their financial situation.

6. Aggravating and Alleviating Factors

The nurse asks the patient what, if anything, makes the pain worse and what makes it better and asks specifically about the relationship between activity and pain. This helps detect factors associated with pain. For example, in a patient with advanced metastatic cancer, pain with coughing may signal spinal cord compression.

The nurse ascertains whether environmental factors influence pain because they may easily be changed to help the patient. For example, making the room warmer may help a patient relax and may decrease the person’s pain.

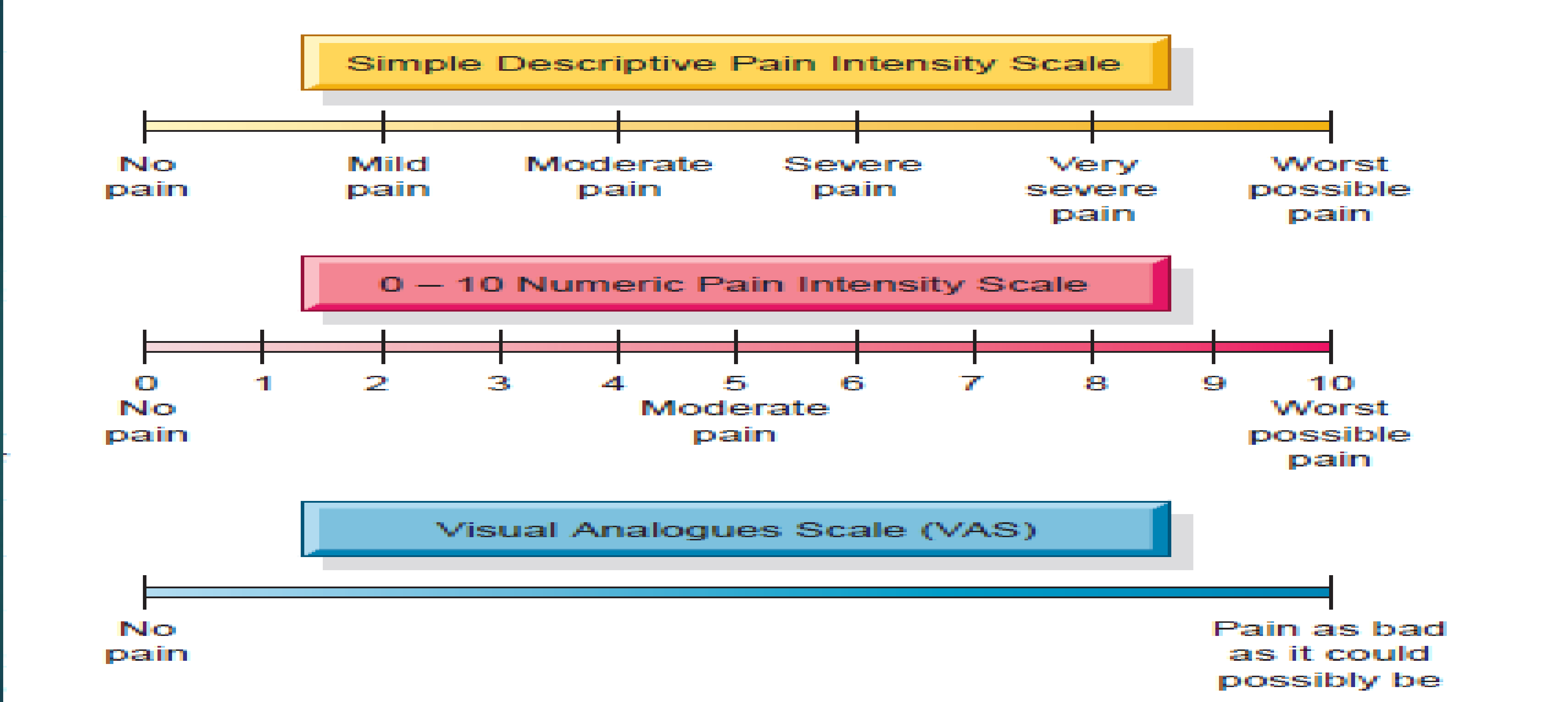
Instruments for Assessing the Perception of Pain

Several pain assessment tools may be used to document the need for intervention, to evaluate the effectiveness of the intervention, and to identify the need for alternative or additional interventions if the initial intervention is ineffective in relieving the pain.

Visual Analogue Scales and Other Intensity Scales

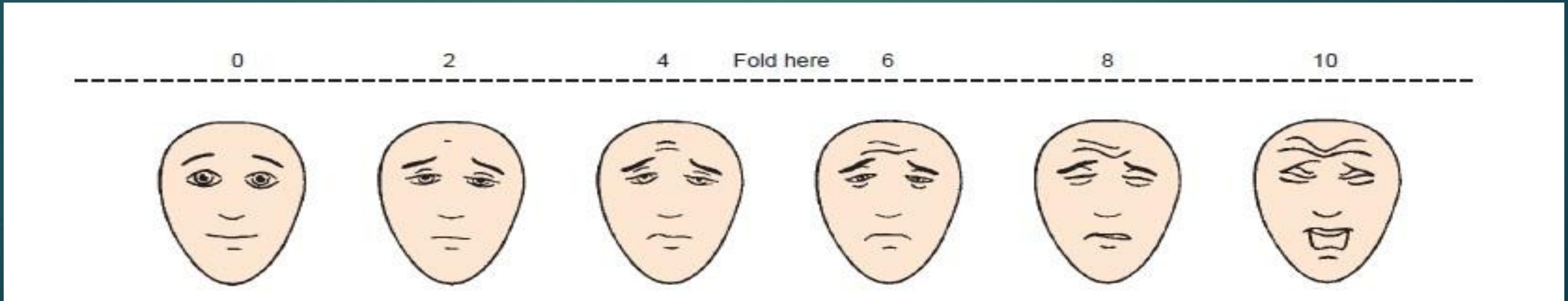
Some patients (eg, children, elderly patients, visually or cognitively impaired patients) may find it difficult to use an unmarked VAS. In those circumstances, ordinal scales, such as a simple descriptive pain intensity scale or a 0-to-10 numeric pain intensity scale, may be used.

Pain intensity scale



Faces Pain Scale–Revised

This instrument has six faces depicting expressions that range from contented to obvious distress. The patient is asked to point to the face that most closely resembles the intensity of his or her pain.



Application of nursing process for patients with pain

Assessment

An accurate assessment focusing on pain cause, which is essential for determining the proper therapy. Ongoing assessment also is important for implementing an effective pain management plan.

II.Diagnosis:

- Diagnostic statement and care plan should identify the following: -

- Type and/or location of pain as postoperative chest pain.
- Etiologic factors and precipitating factors.
- The client's responses

Examples of pain:

- Postoperative pain related to fear of taking prescribed analgesics.
- Chronic Pain: Headache related to inadequate pain management.
- Chronic Pain related to rheumatoid arthritis.

III-Planning: Client Goals

Long-Term Goal

The client will: Verbalize that pain is eliminated.

Short-Term Goals

The client will: Describe a gradual reduction of pain.

IV- Implementing:

A: Establishing a trusting nurse- patient relationship

B: Teaching the client about pain:

- Cause of pain
- Factors increased or decreased.

- Quality and duration of pain to expect.
- Assurance that it is acceptable to express feelings about pain.
- What pain-control measures can and will be used

C-Manipulating Factors Affecting the Pain Experience:

1.Removing or altering the cause of the pain:

- Ways of doing this include:

a. Taking steps to relieve constipation and flatus.

b. Changing body positions and ensuring correct body alignment.

- Changing soiled linens & dressings that maybe irritating the skin.
- The hungry or thirsty.
- Certain drugs are useful for removing or altering the intensity of painful stimuli.

Non pharmacologic management of pain

- ***Massage***

- Several non-pharmacologic pain relief strategies, including rubbing the skin and using heat and cold.

- **Massage**, which is generalized cutaneous stimulation of the body, often concentrates on the back and shoulders. Massage also promotes comfort because it produces muscle relaxation.

Thermal Therapies

Ice and heat therapies may be effective pain relief strategies in some circumstances; however, their effectiveness and mechanisms of action need further study.

For greatest effect, ice should be placed on the injury site immediately after injury or surgery. Ice therapy after joint surgery can significantly reduce the amount of analgesic medication required.

- Care must be taken to assess the skin before treatment and to protect the skin from direct application of the ice.
- Ice should be applied to an area for no longer than 15 to 20 minutes. Long applications of ice may result in frostbite or nerve injury.

- Application of heat increases blood flow to an area and contributes to pain reduction by speeding healing. Both ice and heat therapy must be applied carefully and monitored closely to avoid injuring the skin.

Transcutaneous Electrical Nerve Stimulation

- Transcutaneous electrical nerve stimulation (TENS) uses a battery-operated unit with electrodes applied to the skin to produce a tingling, vibrating, or buzzing sensation in the area of pain. It has been used in both acute and chronic pain relief.
- For example, when TENS is used in a postoperative patient, the electrodes are placed around the surgical wound.

Distraction

- Distraction helps relieve both acute and chronic pain. Distraction, which involves focusing the patient's attention on something other than the pain.
- Distraction techniques may range from simple activities, such as watching TV or listening to music.

- Visits from family and friends can be effective in relieving pain. Watching an action-packed movie may be effective.
- Not all patients obtain pain relief with distraction, especially those in severe pain.
- Severe pain may prevent patients from concentrating well enough to participate in complex physical or mental activities.

- *Relaxation Techniques*

When teaching this technique, the nurse may count out loud with the patient at first. Slow, rhythmic breathing may also be used as a distraction technique.

- ***Guided Imagery***

Guided imagery is using one's imagination to achieve a specific positive effect. Guided imagery for relaxation and pain relief may consist of combining slow, rhythmic breathing with a mental image of relaxation and comfort. The nurse instructs the patient to close both eyes and breathe slowly in and out.

Usually, the patient is asked to practice guided imagery for about 5 minutes, three times a day. Several days of practice may be needed before the intensity of pain is reduced.

- *Hypnosis*

Hypnosis, which has been effective in relieving pain or decreasing the amount of analgesic agents required in patients with acute and chronic pain, may promote pain relief in particularly difficult situations (eg, burns). The mechanism by which hypnosis acts is unclear. Its effectiveness depends on the hypnotic susceptibility of the individual. Usually, hypnosis must be induced by especially skilled people (a psychologist or a nurse with specialized training in hypnosis). Some patients may learn to perform self-hypnosis.

- ***Music Therapy***

Music therapy is an inexpensive and effective therapy for the reduction of pain and anxiety. Research among elderly Korean and American women who had gynecologic surgery demonstrated decreased pain among those patients who received a music therapy intervention.

Alternative Therapies

People suffering chronic, debilitating pain are often desperate. They may try anything, recommended by anyone, at any price. Therapies recommended for pain from these sources include, but are not limited to, therapeutic touch, herbal therapy, reflexology, acupressure, and homeopathy. Many of these “therapies” are probably not harmful.

Nurses should encourages the patient to assess the effectiveness of the therapy, continually using standard pain assessment techniques.

Pharmacologic intervention

(A)Analgesic Administration

- There are three general classes of drugs used for pain relief:
  - Nonnarcotic analgesics (eg, aspirin, nonsteroidal anti-inflammatory agents)
  - Narcotic analgesics or opioids (morphine).

- Adjuvant analgesics (anticonvulsants, antidepressants, and others)

(B) Local Anesthesia: Anesthetic agents may be applied topically to the skin or mucous

- membranes or injected into the body to produce a temporary loss of sensation.

© Neurosurgery

1. **Neurectomy:** is a partial or total excision of a peripheral or cranial nerve to relieve localized pain
2. **Cordotomy** is a surgical resection of the spinal and cerebral tracts carrying painful impulses and is used for advanced disease pain

3.Sympathectomy is a severing of the sympathetic afferent nerve pathways that is used primarily for casualties, and pain due to vascular disorders.

Test your self

- Types of pain according to origin
- *Manifestations of pain*
- *Factors Influencing Pain Response*